

D870

# Physical Chemistry I

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

## How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A 5 mW laser pointer sold as a classroom demo kit emits green light at 500 nm. What is the energy of one photon? ( $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ ,  $c = 3.00 \times 10^8 \text{ m/s}$ )

- A.  $9.94 \times 10^{-32} \text{ J}$
- B.  $1.33 \times 10^{-27} \text{ J}$
- C.  $3.98 \times 10^{-19} \text{ J}$
- D.  $3.98 \times 10^{-28} \text{ J}$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

2. [TRANSFER — reversed stem / different variable]

A photon carries  $4.97 \times 10^{-19} \text{ J}$ . What is its wavelength? ( $hc = 1.99 \times 10^{-25} \text{ J}\cdot\text{m}$ )

- A. 400 nm
- B.  $4.00 \times 10^{-7} \text{ nm}$
- C. About 800 nm, since this photon has more energy than a 500 nm photon
- D.  $2.50 \times 10^6 \text{ m}$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

3.

A hydrogen gas-discharge tube running at 5,000 V glows pink-red in a darkened room. Which electron transition emits the highest-energy photon?

- A.  $n = 3 \rightarrow n = 2$
- B.  $n = 4 \rightarrow n = 1$
- C.  $n = 1 \rightarrow n = 4$
- D.  $n = 4 \rightarrow n = 3$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

4.

A food-coloring manufacturer models a dye's conjugated chain as a particle in a box. A new dye with a brighter shade has a box twice as long. By what factor does the ground-state energy change?

- A. It becomes one-fourth as large.
- B. It becomes one-half as large.
- C. It never changes, because energy levels are fixed for an electron.
- D. It becomes four times as large.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5.

A gas in a steel cylinder with a greased piston, held at 2.0 atm, absorbs 500 J of heat and does 200 J of work pushing the piston outward. What is  $\Delta U$  for the gas?

- A. +300 J
- B. -300 J
- C. +500 J
- D. +700 J

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6. [TRANSFER — reversed stem / different variable]

A system's internal energy falls by 150 J while the surroundings do 250 J of work on it. What is  $q$ ?

- A. -150 J
- B. +100 J
- C. +400 J
- D. -400 J (400 J of heat released)

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

An engineer compares four reactions for a plant report; one of them is run in a kiln at 900 °C. Which reaction has a positive  $\Delta S$ ?

- A.  $\text{H}_2\text{O(g)} \rightarrow \text{H}_2\text{O(l)}$
- B.  $\text{CaCO}_3\text{(s)} \rightarrow \text{CaO(s)} + \text{CO}_2\text{(g)}$
- C.  $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightarrow 2\text{NH}_3\text{(g)}$
- D.  $2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{SO}_3\text{(g)}$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

A drug stored in a pharmacy refrigerator decomposes by first-order kinetics with  $k = 0.0231 \text{ min}^{-1}$  at room temperature. Its initial concentration is 250 mg/L. What is its half-life at room temperature?

- A. 0.0160 min
- B. 43.3 min
- C. 30.0 min
- D. Longer than 30 min, because a larger starting amount takes longer to halve

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9. [TRANSFER — reversed stem / different variable]

A first-order reactant has a half-life of 12 h. What fraction remains after 36 h?

- A. None; three half-lives completes the reaction
- B.  $1/8$  (12.5%)
- C.  $1/4$  (25%)
- D.  $1/6$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

A dairy that ships to three states stores milk at 4 °C instead of 20 °C to slow spoilage reactions. Its cooling system was upgraded last year. Why does the higher temperature speed the reactions?

- A. The rate always exactly doubles for every 10 °C rise.
- B. The higher temperature lowers the activation energy.
- C. The higher temperature makes  $\Delta H$  more negative.
- D. A larger fraction of collisions have energy at or above the activation energy.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11. [TRANSFER — reversed stem / different variable]

Reaction A has  $E_a = 40 \text{ kJ/mol}$  and reaction B has  $E_a = 100 \text{ kJ/mol}$ . Both are warmed by the same 10 K. Which rate constant increases by the larger factor?

- A. It cannot be judged without their  $\Delta H$  values
- B. Both increase by the same factor, since the temperature change is the same
- C. B, the one with the higher activation energy
- D. A, because it is already the faster reaction

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**12.**

At 500 K, the reaction  $\text{NO}_2 + \text{CO} \rightarrow \text{NO} + \text{CO}_2$  goes by two steps: Step 1 (slow):  $\text{NO}_2 + \text{NO}_2 \rightarrow \text{NO}_3 + \text{NO}$ . Step 2 (fast):  $\text{NO}_3 + \text{CO} \rightarrow \text{NO}_2 + \text{CO}_2$ . CO is a toxic gas monitored in the lab. What is the rate law?

**A.** Rate =  $k[\text{NO}_2]^2[\text{CO}]$

**B.** Rate =  $k[\text{NO}_3][\text{CO}]$

**C.** Rate =  $k[\text{NO}_2][\text{CO}]$

**D.** Rate =  $k[\text{NO}_2]^2$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer C Concept 1

A ~~5-mW~~ laser pointer ~~sold as a classroom demo kit~~ emits green light at 500 nm. What is the energy of one photon? ( $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ ,  $c = 3.00 \times 10^8 \text{ m/s}$ )

**Target word:** energy per photon **Decides:**  $E = hc/\lambda$  with  $\lambda = 5.00 \times 10^{-7} \text{ m}$ .

|   |                    |                                                                                                            |
|---|--------------------|------------------------------------------------------------------------------------------------------------|
| A | Inverted operation | Multiplies by $\lambda$ instead of dividing.                                                               |
| B | Dropped element    | $h/\lambda$ leaves out the speed of light.                                                                 |
| C | <b>Correct</b>     | $(6.626 \times 10^{-34} \times 3.00 \times 10^8) / 5.00 \times 10^{-7} = 3.98 \times 10^{-19} \text{ J}$ . |
| D | Wrong stem number  | Uses 500 instead of converting nm to m.                                                                    |

## 2. Answer A Concept 1 • transfer

A photon carries  $4.97 \times 10^{-19} \text{ J}$ . What is its wavelength? ( $hc = 1.99 \times 10^{-25} \text{ J}\cdot\text{m}$ )

**Target word:** solve for wavelength **Decides:**  $\lambda = hc/E$ .

|   |                    |                                                                                                  |
|---|--------------------|--------------------------------------------------------------------------------------------------|
| A | <b>Correct</b>     | $1.99 \times 10^{-25} / 4.97 \times 10^{-19} = 4.00 \times 10^{-7} \text{ m} = 400 \text{ nm}$ . |
| B | Dropped element    | Right number in meters, but labeled nm without converting.                                       |
| C | Confusion set      | More energy means a shorter wavelength, not a longer one.                                        |
| D | Inverted operation | $E/hc$ flips the ratio.                                                                          |

## 3. Answer B Concept 2

A hydrogen gas-discharge tube ~~running at 5,000 V glows pink red in a darkened room~~. Which electron transition emits the highest-energy photon?

**Target word:** largest emitted energy **Decides:** Largest downward gap: to  $n = 1$  from the highest level listed.

|   |                    |                                                                                       |
|---|--------------------|---------------------------------------------------------------------------------------|
| A | Camouflage         | This is the visible red line that matches the glow, but it is not the highest energy. |
| B | <b>Correct</b>     | Levels crowd together at high $n$ ; a drop to $n = 1$ spans the largest gap.          |
| C | Inverted operation | An upward jump absorbs energy; it does not emit a photon.                             |
| D | Confusion set      | Large $n$ values but a tiny gap; this is a low-energy infrared line.                  |

## 4. Answer A Concept 3

A ~~food-coloring manufacturer~~ models a dye's conjugated chain as a particle in a box. A new dye ~~with a brighter shade~~ has a box twice as long. By what factor does the ground-state energy change?

**Target word:** energy scaling with box length **Decides:**  $E \propto 1/L^2$ ; doubling  $L$  gives  $1/2^2$ .

|   |                    |                                                    |
|---|--------------------|----------------------------------------------------|
| A | <b>Correct</b>     | $1 / 2^2 = 1/4$ .                                  |
| B | Dropped element    | Leaves out the square on $L$ .                     |
| C | Absolute language  | Levels depend on the size of the confining region. |
| D | Inverted operation | Energy falls, not rises, as the box lengthens.     |

### 5. Answer A Concept 4

A gas in a steel cylinder ~~with a greased piston, held at 2.0 atm,~~ absorbs 500 J of heat and does 200 J of work pushing the piston outward. What is  $\Delta U$  for the gas?

**Target word:** change in internal energy **Decides:**  $q = +500$  J; work done by the system  $w = -200$  J.

|   |                    |                                                     |
|---|--------------------|-----------------------------------------------------|
| A | <b>Correct</b>     | $\Delta U = 500 + (-200) = +300$ J.                 |
| B | Confusion set      | Flips both signs.                                   |
| C | Dropped element    | Ignores the work term.                              |
| D | Inverted operation | Adds work done by the gas as if it were done on it. |

### 6. Answer D Concept 4 • transfer

A system's internal energy falls by 150 J while the surroundings do 250 J of work on it. What is  $q$ ?

**Target word:** solve for  $q$  **Decides:**  $q = \Delta U - w$  with  $\Delta U = -150$  J and  $w = +250$  J.

|   |                    |                                                    |
|---|--------------------|----------------------------------------------------|
| A | Wrong stem number  | Repeats $\Delta U$ .                               |
| B | Inverted operation | Adds $w$ to $\Delta U$ instead of subtracting.     |
| C | Confusion set      | Right size, wrong sign; the system must shed heat. |
| D | <b>Correct</b>     | $-150 - 250 = -400$ J.                             |

### 7. Answer B Concept 5

An engineer compares four reactions for a plant report; ~~one of them is run in a kiln at 900 °C.~~ Which reaction has a positive  $\Delta S$ ?

**Target word:** sign of  $\Delta S$  **Decides:** The reaction that makes gas from a solid.

|   |                    |                                                          |
|---|--------------------|----------------------------------------------------------|
| A | Adjacent category  | Condensation moves toward order; $\Delta S$ is negative. |
| B | <b>Correct</b>     | Zero moles of gas become one; disorder rises.            |
| C | Inverted operation | Four moles of gas become two; $\Delta S$ is negative.    |
| D | Confusion set      | Three moles of gas become two; $\Delta S$ is negative.   |

### 8. Answer C Concept 6

A drug ~~stored in a pharmacy refrigerator~~ decomposes by first-order kinetics with  $k = 0.0231 \text{ min}^{-1}$  at room temperature. ~~Its initial concentration is 250 mg/L.~~ What is its half-life at room temperature?

**Target word:** first-order half-life **Decides:**  $t_{1/2} = 0.693 / k$ .

|   |                    |                                                                  |
|---|--------------------|------------------------------------------------------------------|
| A | Inverted operation | Multiplies 0.693 by $k$ .                                        |
| B | Dropped element    | $1 / k$ leaves out $\ln 2$ .                                     |
| C | <b>Correct</b>     | $0.693 / 0.0231 = 30.0$ min.                                     |
| D | Camouflage         | First-order half-life does not depend on starting concentration. |

### 9. Answer B Concept 6 • transfer

A first-order reactant has a half-life of 12 h. What fraction remains after 36 h?

**Target word:** fraction remaining **Decides:**  $36 / 12 = 3$  half-lives;  $(\frac{1}{2})^3$ .

|          |                    |                                                                                 |
|----------|--------------------|---------------------------------------------------------------------------------|
| A        | Absolute language  | First-order decay never reaches exactly zero.                                   |
| <b>B</b> | <b>Correct</b>     | $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 1/8$ .                     |
| C        | Wrong stem number  | Counts two half-lives.                                                          |
| D        | Inverted operation | Multiplies $\frac{1}{2}$ by 3 half-lives instead of raising to the third power. |

### 10. Answer D Concept 7

A dairy ~~that ships to three states~~ stores milk at 4 °C instead of 20 °C to slow spoilage reactions. ~~Its cooling system was upgraded last year.~~ Why does the higher temperature speed the reactions?

**Target word:** mechanism of the temperature effect **Decides:**  $E_a$  is unchanged; more molecules have enough energy to cross it.

|          |                   |                                                                   |
|----------|-------------------|-------------------------------------------------------------------|
| A        | Absolute language | That is a rough rule of thumb; the factor depends on $E_a$ .      |
| B        | Confusion set     | Lowering $E_a$ is what a catalyst does.                           |
| C        | Adjacent category | $\Delta H$ is a thermodynamic quantity; it does not set the rate. |
| <b>D</b> | <b>Correct</b>    | The Boltzmann distribution shifts; $k$ rises exponentially.       |

### 11. Answer C Concept 7 • transfer

Reaction A has  $E_a = 40$  kJ/mol and reaction B has  $E_a = 100$  kJ/mol. Both are warmed by the same 10 K. Which rate constant increases by the larger factor?

**Target word:** which reaction is more temperature-sensitive **Decides:** The Arrhenius slope is  $-E_a/R$ ; larger  $E_a$  means a steeper response.

|          |                    |                                                       |
|----------|--------------------|-------------------------------------------------------|
| A        | Adjacent category  | $\Delta H$ does not appear in the Arrhenius equation. |
| B        | Absolute language  | The factor depends on $E_a$ .                         |
| <b>C</b> | <b>Correct</b>     | Higher $E_a$ makes $k$ more sensitive to temperature. |
| D        | Inverted operation | Lower $E_a$ gives a smaller relative increase.        |

### 12. Answer D Concept 8

~~At 500 K,~~ the reaction  $\text{NO}_2 + \text{CO} \rightarrow \text{NO} + \text{CO}_2$  goes by two steps: Step 1 (slow):  $\text{NO}_2 + \text{NO}_2 \rightarrow \text{NO}_3 + \text{NO}$ . Step 2 (fast):  $\text{NO}_3 + \text{CO} \rightarrow \text{NO}_2 + \text{CO}_2$ . ~~CO is a toxic gas monitored in the lab.~~ What is the rate law?

**Target word:** rate law **Decides:** The slow step involves two  $\text{NO}_2$ .

|          |                   |                                                                        |
|----------|-------------------|------------------------------------------------------------------------|
| A        | Camouflage        | Pulled in by the attention on CO; CO does not appear in the slow step. |
| B        | Adjacent category | That is the fast step, and $\text{NO}_3$ is an intermediate.           |
| C        | Confusion set     | Written from the overall equation, not the slow step.                  |
| <b>D</b> | <b>Correct</b>    | The rate-determining step is bimolecular in $\text{NO}_2$ .            |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 5             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 3             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 4             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 10            |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 4             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 7             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 3             |             |

## Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                             | Rule                                                                                                                                                                        | Trap                                                                                          | Tell                                                                    |
|---|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1 | <b>Photon energy and wavelength</b>                 | $E = hc/\lambda$ : energy is inversely proportional to wavelength; $\lambda$ must be in meters.                                                                             | Treating energy as proportional to wavelength or leaving $\lambda$ in nanometers.             | A wavelength in nm and a request for energy in joules (or the reverse). |
| 2 | <b>Hydrogen emission transitions</b>                | Emission is a drop to a lower level; the larger the energy gap, the higher the photon energy and the shorter the wavelength. Transitions to $n = 1$ are the most energetic. | Judging energy by the size of the $n$ numbers rather than the gap.                            | A list of $n \rightarrow n$ transitions.                                |
| 3 | <b>Particle in a box</b>                            | Energy levels scale as $n^2/L^2$ ; a longer box lowers and crowds the levels.                                                                                               | Dropping the square on $L$ or thinking levels do not depend on box size.                      | A conjugated chain or box length that changes.                          |
| 4 | <b>First law sign conventions</b>                   | $\Delta U = q + w$ ; heat into the system and work done on the system are positive, work done by the system is negative.                                                    | Adding work done by the system as a positive term.                                            | "Absorbs heat" and "does work on the surroundings."                     |
| 5 | <b>Predicting the sign of <math>\Delta S</math></b> | Entropy increases when moles of gas increase, solids or liquids become gases, or particles spread out.                                                                      | Counting total moles on each side without checking which are gases.                           | Balanced equations with phase labels.                                   |
| 6 | <b>First-order half-life</b>                        | $t_{1/2} = 0.693/k$ and does not depend on starting concentration; after $n$ half-lives, $(1/2)^n$ remains.                                                                 | Thinking a larger starting amount takes longer to halve, or dividing instead of using powers. | A first-order rate constant or a count of half-lives.                   |
| 7 | <b>Temperature and the Arrhenius equation</b>       | Raising temperature increases $k$ because more collisions exceed $E_a$ ; $E_a$ itself does not change, and reactions with higher $E_a$ are more temperature-sensitive.      | Saying heat lowers the activation energy (that is what a catalyst does).                      | A temperature change with no catalyst.                                  |
| 8 | <b>Rate law from the slow step</b>                  | The rate law comes from the molecularity of the rate-determining (slow) step, not the overall equation.                                                                     | Writing the rate law from overall stoichiometry.                                              | A multistep mechanism with one step marked slow.                        |

**Course scope (public description):** Physical Chemistry I provides an exploration of the intersection of physics and chemistry to understand the fundamental principles that govern chemical reactions and matter. This course covers key topics such as atomic and molecular structure, thermodynamics, and kinetics, emphasizing real-world applications and practical problem-solving. Through a blend of virtual labs and interactive activities, learners will develop a deep understanding of energy changes, reaction mechanisms, and the behavior of matter in various states. This course emphasizes engagement with complex concepts, enhancement of analytical skills, and application of theoretical knowledge to practical scenarios.